

YUBO LI

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EDUCATION

Carnegie Mellon University - <i>Ph.D. in Information Systems</i>	2022 – Present
Carnegie Mellon University - <i>M.S. in Information Systems & Management</i>	2020 – 2021
University of California, San Diego - <i>B.S. in Applied Mathematics Minor in Business</i>	2015 – 2019
<i>Leadership: Founder of the UCSD Triton Actuarial Society GPA: 3.8/4.0</i>	

SELECTED PUBLICATIONS

- [Consistency of Large Reasoning Models Under Multi-Turn Attacks.](#)
- [The Model Says Walk: How Surface Heuristics Override Implicit Constraints in LLM Reasoning.](#)
- [Skills-Bench: Benchmarking the Efficacy of Agent Skills Across Diverse Tasks](#), NeurIPS 2026
- [Beyond Single-Turn: A Survey on LLM Multi-Turn Interactions.](#) TMLR 2026.
- [Dynamic Agent Skills: A Lifecycle Survey and Taxonomy of Evolving Skill Libraries.](#) TMLR 2026.
- [Time-To-Inconsistency: A Survival Analysis of Large Language Model Robustness to Adversarial Attacks.](#) ICLR 2026.
- [Firm or Fickle? Evaluating LLM Consistency Capability in Multi-Turn Interactions.](#) ACL 2025.
- [No Black Box Anymore: Demystifying Clinical Predictive Modeling with Temporal-Feature Cross Attention Mechanism.](#) American Medical Informatics Association 2025 Annual Symposium (3rd Place in Paper Competition).
- [Enhancing End Stage Renal Disease Outcome Prediction: A Multi-Sourced Data-Driven Approach.](#) Journal of the American Medical Informatics Association, 2025.
- [Co-Alignment: Rethinking Alignment as Bidirectional Human-AI Cognitive Adaptation.](#) Agents4Science 2025.
- [Towards Interpretable End-Stage Renal Disease \(ESRD\) Prediction: Utilizing Administrative Claims Data with Explainable AI Techniques.](#) American Medical Informatics Association 2024 Annual Symposium.

AWARDS & FELLOWSHIPS

- 2026 OpenAI Researcher Access Program
- 2026 Presidential Fellowship, Carnegie Mellon University
- 2025 Presidential Fellowship, Tata Consultancy Services (TCS)
- [2024 Fellowship in Generative AI in Healthcare](#), Center for Machine Learning and Health, Carnegie Mellon University
- [2023 Fellowship in Digital Health Innovation](#), Center for Machine Learning and Health, Carnegie Mellon University

WORKING EXPERIENCE

- Netflix** – *ML Research Scientist Intern* 2026.06 – 2026.08
- **Personalized GenRec:** Developed internal personalized foundational model that jointly encode free-text queries and user interaction history to produce ranked semantic-ID recommendation lists, improving MRR@50 from 0.22 to 0.67.
 - **Spectral Analysis & Capability Diagnostics:** Developed a label-free diagnostic framework for post-training forgetting using layerwise CKA, covariance effective rank, and principal-subspace overlap; identified distinct geometric signatures associated with 86% vs. 15% recovery of lost QA accuracy under matched replay training.
 - **Capability Recovery:** Designed and implemented weight-delta interpolation, replay-based fine-tuning, and on-policy distillation (OPD) to mitigate post-training regressions, balancing recovery of legacy capabilities with retention of newly acquired search performance.
 - **Causal Evaluation & Robustness:** Proposed counterfactual evaluation protocols to disentangle whether models learn to use user history causally versus memorizing history patterns, and stress-tested robustness under noisy-history interventions.
- Amazon** – *Applied Scientist Intern* 2025.06 – 2025.08
- **Generative RecSys:** Developed User-LLM pipeline for personalized recommendation by pre-training custom user encoder on 172k customer interaction sequences, then fine-tuning through progressive alignment strategies (adapter-only, adapter + LLM, full fine-tuning) and pooling mechanisms.
 - Achieved superior predictive performance (60.1% Hit Ratio@3, 67.5% NDCG@3) compared to direct prompting baselines while delivering **93.4% token reduction** and **64.8% cost savings**.
 - **RAG/Search:** Architected production-ready RAG service infrastructure using FAISS HNSW indexing for 60k user embeddings to enhance recommendation contest at scale.
 - **Distributed Training:** Engineered distributed training across multi-GPUs using FSDP with mixed-precision BF16.
- R-eat (Self-employed)** - *Founder* 2012 – 2014
- **Entrepreneurship:** Founded and grew an online food delivery business by enhancing customer experience through UI optimization and personalized marketing, resulting in a 60% sales increase and a loyal customer base of over 2,000.

RESEARCH EXPERIENCE

- **Continual Skill Learning for Efficient LLM Agents** **Carnegie Mellon University**
2025.08 - Present
Supervisor: [Dawn Song](#) UC Berkley Professor | [Daniel Fried](#), CMU Professor
 - **Multi-Modal Agents:** Developed PANDO, a single-rollout multimodal web agent that distills interaction trajectories into reusable executable skills; achieved 58.3% success across 910 VisualWebArena tasks, outperforming SGV by 4.3 percentage points with 58% fewer tokens and no pre-evaluation discovery budget.
 - **SkillsBench:** Contributed to SkillsBench, a containerized agent-skills benchmark spanning 87 tasks, 8 domains, and 18 model-harness configurations; paired evaluations with deterministic verifiers showed a 16.6-percentage-point mean pass-rate gain from curated skills.
 - **Dynamic Skills:** Lead a TMLR survey, developing an eight-stage lifecycle framework for evolving skill libraries and synthesizing evidence on verification, retrieval scaling, maintenance, and governance into lifecycle-aware evaluation recommendations..
- **Multi-Turn Robustness & Consistency of LLMs under Attacks** **Carnegie Mellon University**
2025.01 - Present
Supervisor: [Ramayya Krishnan](#), Dean of the Heinz College | [Rema Padman](#), CMU Professor
 - **Evaluation:** Proposed a metric to quantitatively assess multi-turn response stability, prioritizing early interaction accuracy and swift recovery from initial errors. Released a comprehensive benchmark to test the model's ability to sustain coherent and contextually accurate responses during dynamic, multi-turn conversations.
 - **Failure Prediction:** Formulated time-to-inconsistency as a survival-analysis problem over 36,951 turns from nine LLMs, incorporating semantic drift into failure-risk prediction; developed an AFT-based early-warning monitor that flagged 76% of failing dialogues with a two-turn median lead time in held-out retrospective evaluation.
 - **Mitigation Solution:** Developed **Confidence-Aware Response Generation (CARG)**, an inference-time framework that embeds log-probability-derived confidence into conversation history; achieved 74.82% mean follow-up accuracy versus 71.34% for the GPT-4o baseline, while reducing across-turn accuracy standard deviation by 63%.
 - **Distributed Training:** Architected and optimized distributed training infrastructure for LLMs (Mistral, LLaMA, Deepseek) across 4 nodes with 16 NVIDIA GH200 GPUs, implementing tensor parallelism, pipeline parallelism, and ZeRO optimizer techniques to efficiently scale model training and reduce computational overhead.
- **Q&A Systems for Trustworthy Organ Transplantation Guidance** **Carnegie Mellon University**
2025.01 - Present
Supervisor: [Ramayya Krishnan](#), Dean of the Heinz College | [Rema Padman](#), CMU Professor
 - **Q&A Systems:** Developed Q&A systems for organ transplantation patient care by integrating retrieval-augmented generation (RAG) with adaptive frameworks that synthesize patient-specific medical history, clinical guidelines, and dynamic risk factors to deliver personalized guidance and support.
 - **RAG/Retrieval:** Engineered a hybrid retrieval pipeline combining both sparse and dense retrievers (DPR, ANCE) to accurately extract and verify medical content from over 100 transplant-specific handbooks.
 - **LLM Reasoning:** Developed an incentivized reasoning mechanism that align LLM behaviors by rewarding multi-step chain-of-question sequences, enabling the system to reason freely and proactively request critical diagnostic information.
- **Chronic Kidney Disease Progression Analysis with DL & XAI** **Carnegie Mellon University**
2022.07 – 2025.08
Supervisor: [Rema Padman](#), CMU Professor | [UPMC & Highmark](#)
 - **Deep Learning:** Developed Temporal-Feature Cross Attention Mechanism (TFCAM), an attention-based deep learning framework that explicitly captures temporal and feature-level interactions to enhance clinical predictive modeling for Chronic Kidney Disease (CKD) progression.
 - **XAI:** Provided multi-level explainability including temporal insights, feature importance ranking, and cross-temporal feature interactions, enabling clinicians to interpret model predictions transparently and identify actionable clinical insights.

SELECTED ACADEMIC ACTIVITIES

- **Invited Talk & Guest Lecture**
 - POMS 2026 | Invited Tutorial on LLM/AI Agent Measurement Science & Evaluations
 - GenAI in Healthcare | 94-706 Healthcare Information Systems | Spring 26
 - Coding Agents | 95-891 Introduction to AI | Spring 26
 - AI Agents | 95-891 Introduction to AI | Fall 25/Spring 26
 - Deep Learning for NLP | 95-891 Introduction to AI | Spring 25
- **Teaching Assistant**
 - 11-711 Advanced NLP with Prof. Graham Neubig.
 - 10-680/10-681 Mathematical & Computational Foundation for Machine Learning with Prof. Henry Chai.
 - 95-891 Introduction to Artificial Intelligence with Prof. David Steier.
 - 95-865 Unstructured Data Analysis with Prof. George Chen.